

Himasriya Engu

Orlando, FL | himasriyaengu02@gmail.com | +1 (407) 435-7289 | LinkedIn | GitHub | Portfolio

SUMMARY

AI/ML Engineer and Computer Vision Researcher pursuing an M.S. in Computer Science at UCF (GPA: 3.96). Peer-reviewed published researcher with hands-on experience in deep learning, medical image analysis, computer vision, predictive modeling, and exploratory data analysis. Proficient in Python, SQL, PyTorch, TensorFlow, scikit-learn, Power BI, and cloud-based ML deployment. Teaching Assistant at UCF with strong foundations in algorithms, data structures, and statistical problem-solving. Seeking internship and research roles in Machine Learning, Data Analytics, Computer Vision, and Medical AI.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, R, JavaScript, MATLAB

AI / ML: PyTorch, TensorFlow, CNNs, scikit-learn, XGBoost, Transfer Learning, ResNet, VGG, ViT, YOLOv8, Image Segmentation, Object Detection, Model Fine-Tuning, Data Augmentation, LLMs, RAG, LangChain

Data Analytics: Pandas, NumPy, Statistical Analysis, Hypothesis Testing, EDA, Feature Engineering, SHAP, A/B Testing, Predictive Modeling, Data Cleaning, Data Wrangling, Regression, Classification

Data Visualization: Power BI, Tableau, Excel (Pivot Tables, VLOOKUP), Matplotlib, Seaborn, Plotly, Streamlit, Jupyter Notebook, Dashboard Design

Web Technologies: React, HTML, CSS, Flask, Django, FastAPI, Express.js, REST APIs

Databases: MySQL, PostgreSQL, MongoDB, SQL Query Optimization

Cloud / MLOps: AWS (EC2, S3), Docker, MLflow, ChromaDB, Git, Linux, CI/CD

PROFESSIONAL EXPERIENCE

Teaching Assistant

Feb 2026 – Present

University of Central Florida

Orlando, FL

- COP 3502C: Computer Science I — Support 30+ undergraduate students weekly during programming lab sessions and office hours, reinforcing core CS concepts including data structures, algorithms, and Java.
- Grade weekly assignments for a class of 80+ students, providing structured feedback on code quality, logic, and algorithmic correctness; reduced average resubmission rate through targeted rubric revisions.
- Collaborated with instructor to design lab exercises and align grading rubrics, ensuring consistent, fair course delivery across multiple lab sections.
- Proactively debugged student code across topics including recursion, pointers, Big O complexity, and object-oriented design, improving lab completion rates.

Data Science and Machine Learning Intern

May 2024 – Jun 2024

EISystems

Remote

- Conducted end-to-end EDA and statistical analysis on 890+ records, identifying key survival predictors through correlation analysis, distribution plots, and SHAP-based feature importance ranking.
- Performed data cleaning, missing value imputation (median/mode/KNN), and feature engineering to prepare structured datasets; reduced data inconsistency rate by ~30%.
- Built a Titanic Survival Prediction model using Logistic Regression, Random Forest, and SVM; selected best-performing model via cross-validation achieving 82%+ ROC-AUC on hold-out test set.
- Created data visualizations and interactive dashboards using Matplotlib and Seaborn to communicate model results and business insights to non-technical stakeholders.

Full Stack Web Development Intern

Feb 2024 – Apr 2024

DevTown

Remote

- Developed and deployed 3 full-stack web applications using Flask, Django, and Express.js with RESTful API integration and responsive UI design, serving 200+ test users.
- Wrote optimized SQL queries for data retrieval, aggregation, and reporting workflows across MySQL and MongoDB databases, reducing average query response time by 25%.
- Deployed applications on AWS EC2 with JWT-based authentication; implemented S3 storage for user-uploaded assets, improving scalability and session security.
- Participated in Agile sprint planning, pull request code reviews, and iterative feature delivery within a cross-functional team of 6 engineers.

PROJECTS & PUBLICATIONS

Eye Deep-Net: Retinal Disease Diagnosis | *PyTorch, TensorFlow, CNNs, OpenCV, ResNet, VGG* | [GitHub] | [Publication]

- Published in *TURCOMAT*, 15(3), Sep 2024 (Scopus-indexed).
- Designed and trained CNN architectures with transfer learning (ResNet, VGG) for 4-class retinal disease classification (DR, MH, ODR, Normal), achieving ~95% accuracy across 3,000+ fundus images.

- Applied image segmentation and preprocessing including resizing, normalization, contrast enhancement (CLAHE), and data augmentation to improve generalization on imbalanced classes.
- Addressed class imbalance using weighted cross-entropy loss and oversampling (SMOTE); evaluated models with precision, recall, F1-score, and confusion matrices — outperformed baseline CNN by 8%.
- Contributed to experimental design, statistical result analysis, and peer-reviewed manuscript preparation; accepted in TURCOMAT, a Scopus-indexed journal.

Alzheimer's Disease Detection using CNN | *Python, CNN, TensorFlow, MRI Imaging, scikit-learn* | [GitHub]

- Engineered a CNN-based Alzheimer's detection pipeline on MRI brain scans with hippocampus-focused analysis across 4 disease stages (non-demented, mild, moderate, severe), achieving 91%+ classification accuracy.
- Applied MRI preprocessing including skull stripping, intensity normalization, and slice selection; selected optimal architecture based on sensitivity-specificity tradeoffs for clinical applicability.
- Evaluated model performance using accuracy, AUC, and confusion matrix; benchmarked 3 CNN architectures to identify best-performing configuration for early-stage diagnosis.

Real-Time Vehicle Counting and Classification System | *Python, YOLOv8, OpenCV, Computer Vision*

- Developed a real-time computer vision pipeline for vehicle detection, counting, and multi-class classification from live video feeds using YOLOv8, achieving 90%+ detection accuracy across 5 vehicle classes.
- Processed video at 25+ FPS using OpenCV with optimized inference pipeline; tuned NMS and confidence thresholds to reduce false positives by 15% in high-density traffic scenes.
- Evaluated detection performance using mAP and IoU metrics under varying lighting and occlusion conditions.

Hybrid Movie Recommendation System | *Python, scikit-learn, Pandas, Collaborative Filtering, NLP*

- Combined collaborative filtering (matrix factorization via SVD) with content-based filtering using TF-IDF features to generate personalized recommendations, improving hit rate by 12% over single-method baseline.
- Addressed cold-start problem for new users via item content features; evaluated system using precision@k, recall@k, and NDCG metrics on MovieLens 100K dataset.

EDUCATION

University of Central Florida

Master of Science in Computer Science | GPA: 3.96 / 4.0

- Coursework: Machine Learning, Deep Learning, Computer Vision, Advanced Algorithms

Orlando, FL

Aug 2025 – Present

Malla Reddy Engineering College for Women

Bachelor of Technology in Computer Science and Engineering | GPA: 3.56 / 4.0 | *First Class with Distinction* Aug 2021 – Apr 2025

- Coursework: Data Structures, OS, Networks, OOP Java, DBMS, AI & ML, Data Mining, Big Data

Hyderabad, India

CERTIFICATIONS

- Oracle Certified Associate — Oracle
- Power BI Data Analytics Certification — Skill Nation
- AWS Cloud Practitioner Essentials — Amazon Web Services
- Python for Everybody — Coursera (University of Michigan)